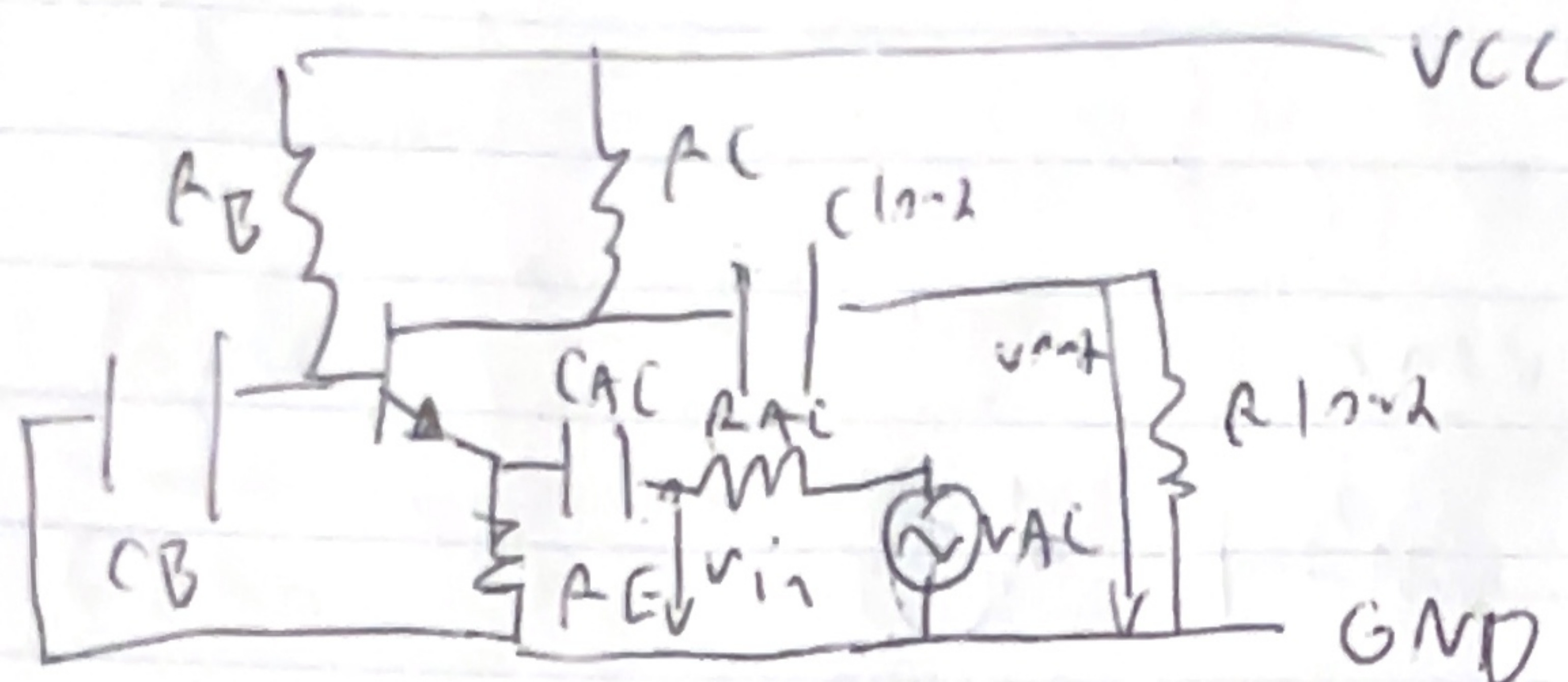
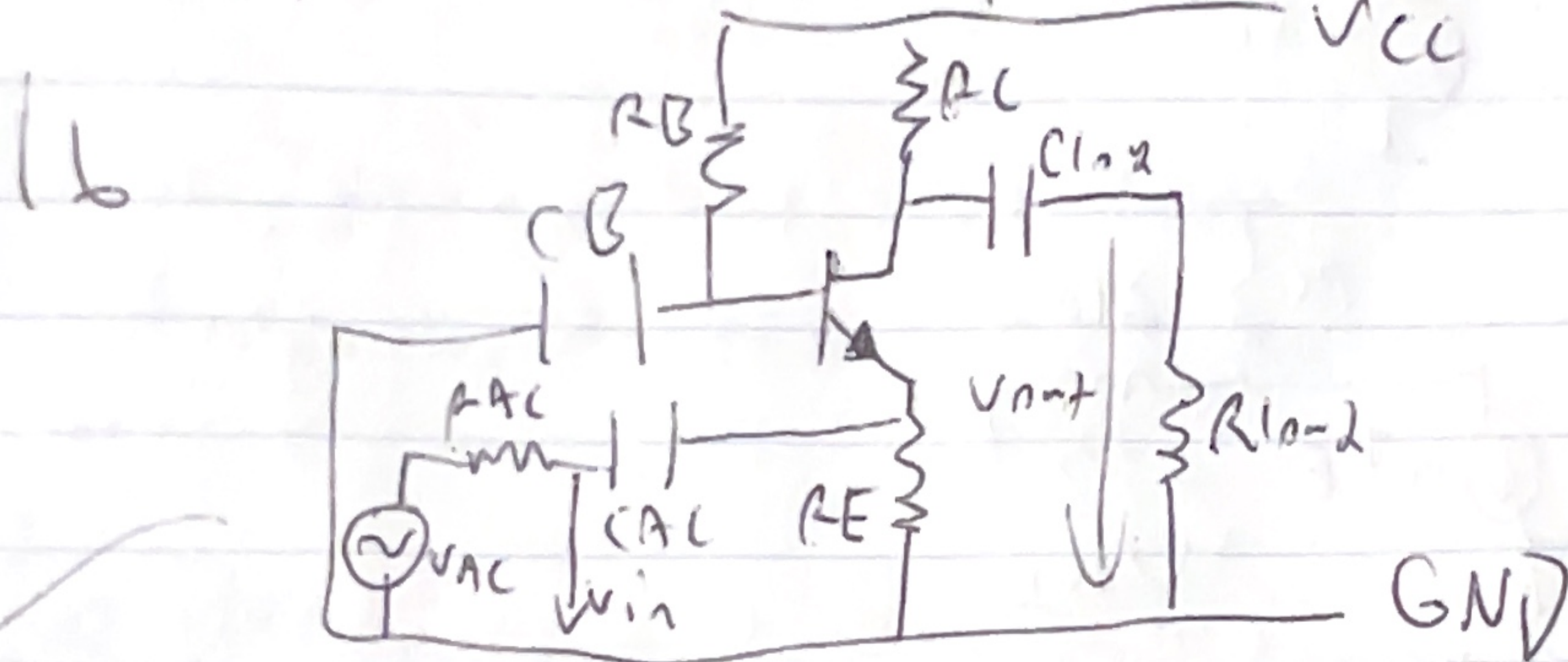


1 $V_{CC} = 12V$ $\alpha = 0.99$ $R_{AC} = 1k$ $R_{load} = 100\Omega$



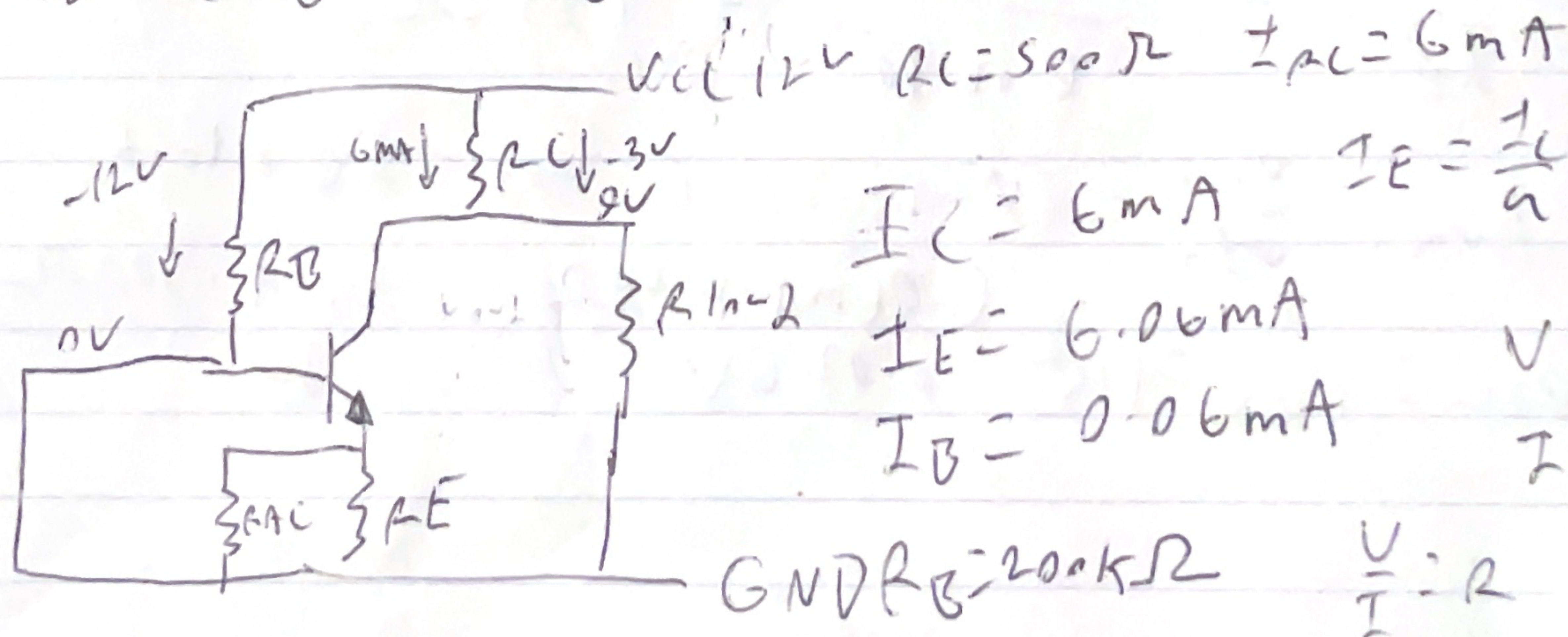
In β , beta. $\beta = \frac{\alpha}{1-\alpha} = \frac{0.99}{1-0.99} = 99$



1c both V_{in} and V_{out} are grounded, and GND connects with base side of the transistor (through a DC blocking capacitor). Since base is grounded, input is through emitter, and output is through collector, this circuit is in common base form.

2 $R_C = R_E = 500\Omega$ $V_{AC} = 3V$ $I_E, I_C, I_B, R_B = ?$

$I_E = I_C + I_B$ $I_C = \alpha I_E$



$V_{CC} = 12V$ $R_C = 500\Omega$ $I_{AC} = 6mA$
 $I_C = 6mA$ $I_E = \frac{I_C}{\alpha}$

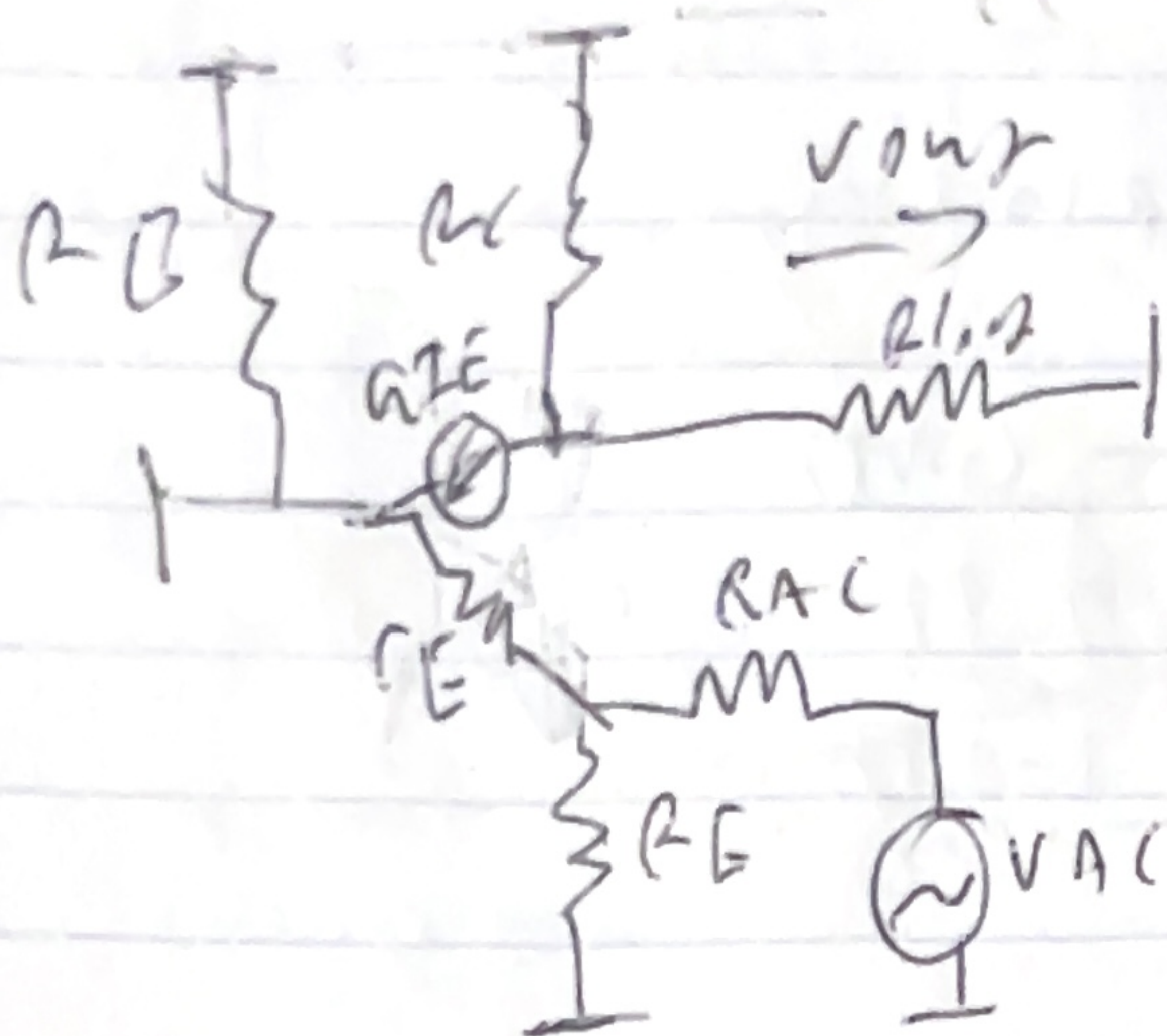
$I_E = 6.06mA$
 $I_B = 0.06mA$

$V_{RB} = 12V$
 $I_{RB} = I_B = 0.06mA$

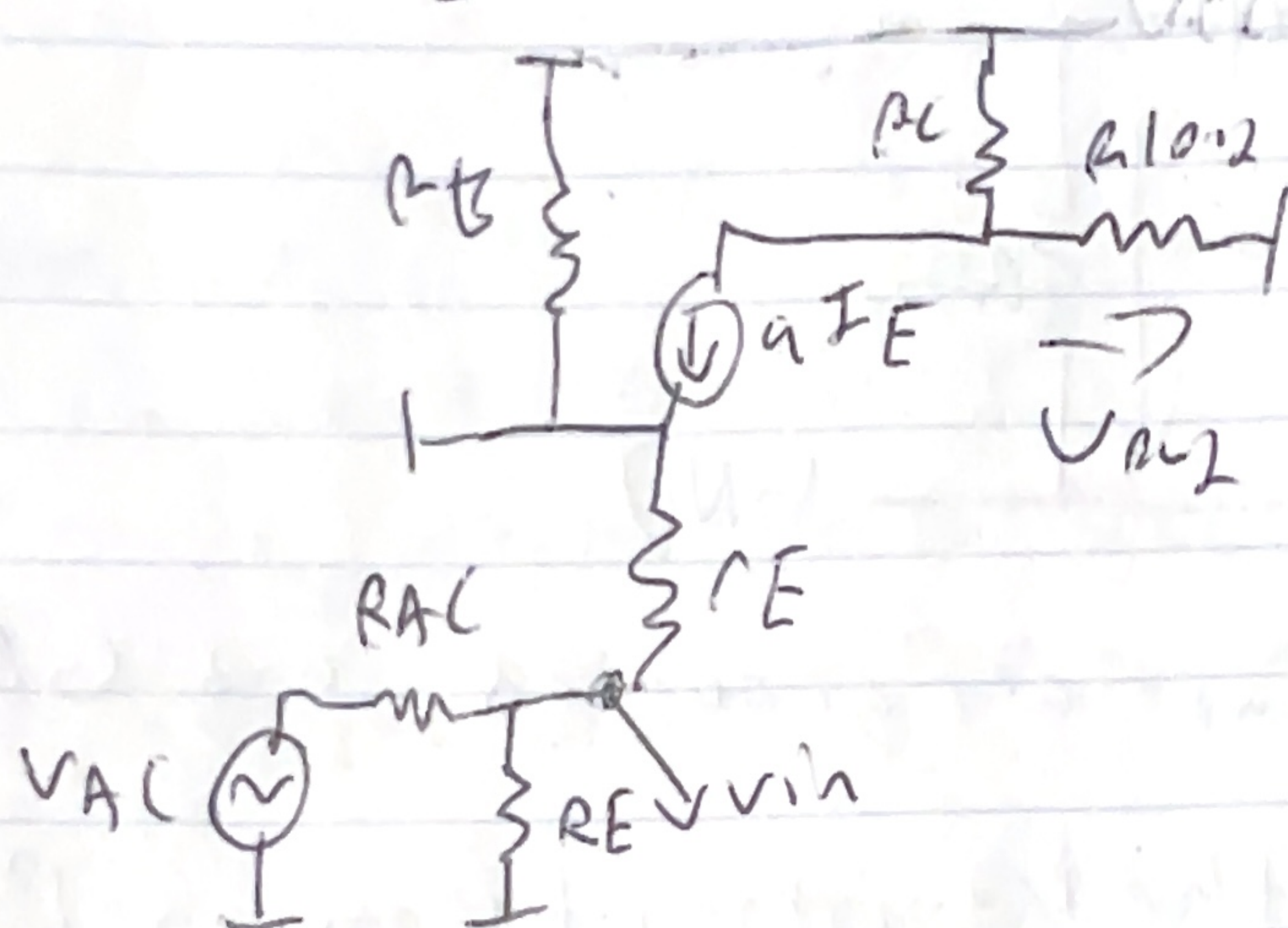
$GND R_B = 200k\Omega$ $\frac{V}{I} = R$

$$r_E = \frac{V_T}{I_E} = \frac{26\text{mV}}{6.06\text{mA}} = 4.29\Omega$$

1f



1g



1h $Z_{in} = R_E = 4.29\Omega$ Relatively low.

1i $Z_{out} = \frac{V_{out}}{I_{out}} = \frac{V_{out}}{I_{E}} = \frac{V_{out}}{I_E} = \frac{V_{out}}{I_E} = \frac{I_{out} R_{load}}{I_E}$

$$\frac{I_C R_{load}}{I_E} = \frac{6 \cdot 10^{-3} \cdot 500}{6.06 \cdot 10^{-3}} = 495\Omega$$

1j $A_{vdc} = \frac{V_{out}}{V_{in}} \quad V_{in} = I_E r_E, \quad V_{out} = I_C R_{load}$

$$A_{vdc} = \frac{3}{0.026} = 115 \quad \frac{6.06\text{mA} \cdot 4.29\Omega}{0.026\text{V}} \quad \frac{6\text{mA} \cdot 500\Omega}{3\text{V}}$$

$$I_C = \alpha I_E$$

$$1K \quad A_{ISC} = \frac{I_{out}}{I_{in}} = \frac{I_C}{I_E} = \alpha = 0.99$$

$$11 \quad A_{VOL} = \frac{v_{out}}{v_{in}} = \frac{I_C R_{load}}{I_E R_E} = \frac{I_C \cdot \frac{R_{load}}{R_E + R_{load}} \cdot R_{load}}{I_E \cdot R_E}$$

$$\frac{6mA \cdot \frac{500}{500+500} \cdot 500}{6.06mA \cdot 500} = 0.495$$

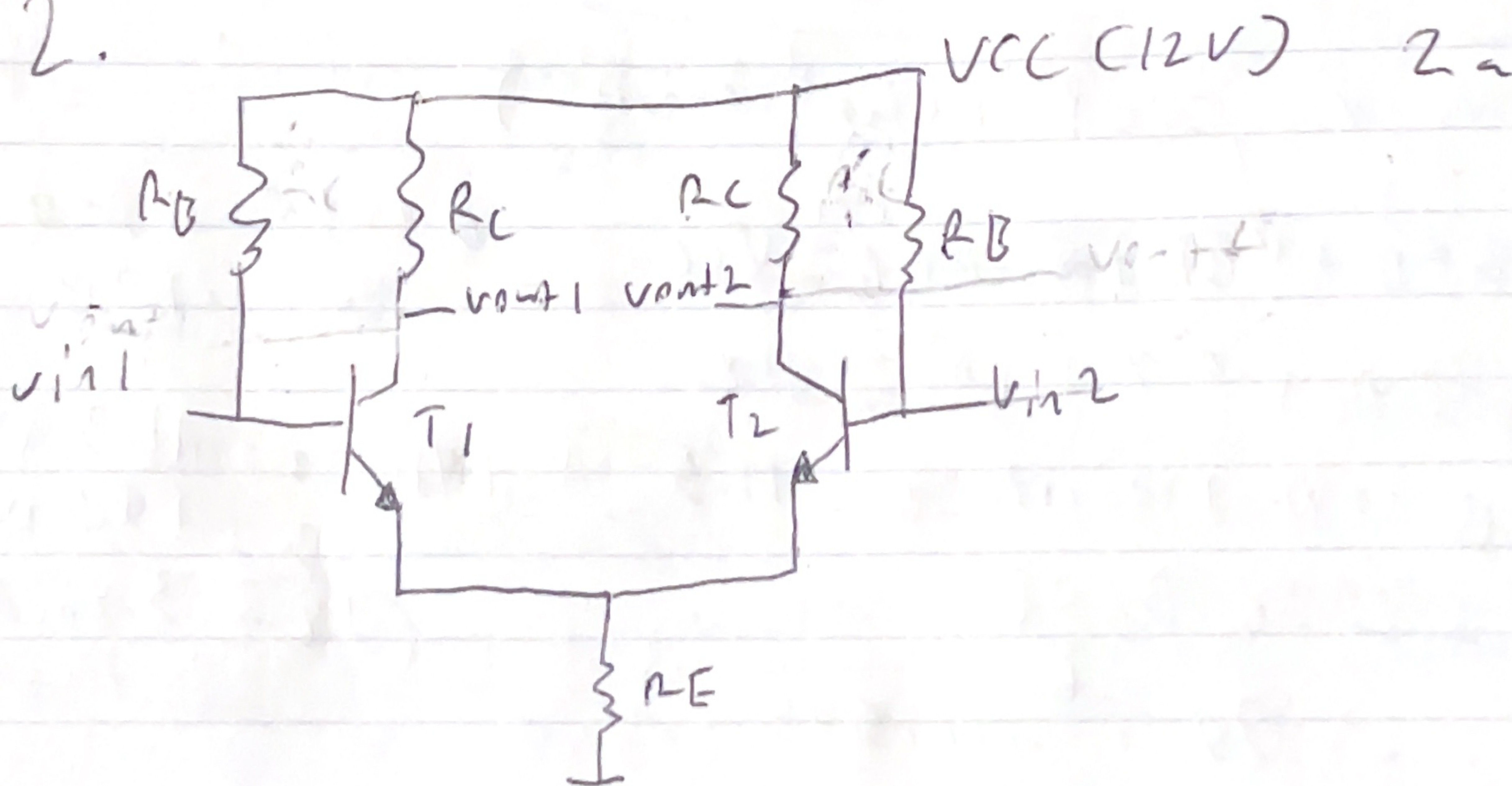
$$1m \quad A_{ISC} = \frac{I_{out}}{I_{in}} = \frac{I_C \cdot \frac{R_{load}}{R_E + R_{load}}}{I_E} = \frac{6mA \cdot \frac{500}{500+500}}{6.06mA} = 0.495$$

It is a high frequency pass filter. The cut off frequency would be $\frac{1}{2\pi RC}$ where $R = R_{load}$ and $C = C_{load}$

$$10 \quad 1000 = \frac{1}{2\pi RC} \quad 2000\pi = \frac{1}{500C} \quad 1000000\pi = \frac{1}{C}$$

$$C = \frac{1}{1000000\pi} \quad C = 3.183 \cdot 10^{-7} F \quad \text{or} \quad 0.3183 \mu F$$

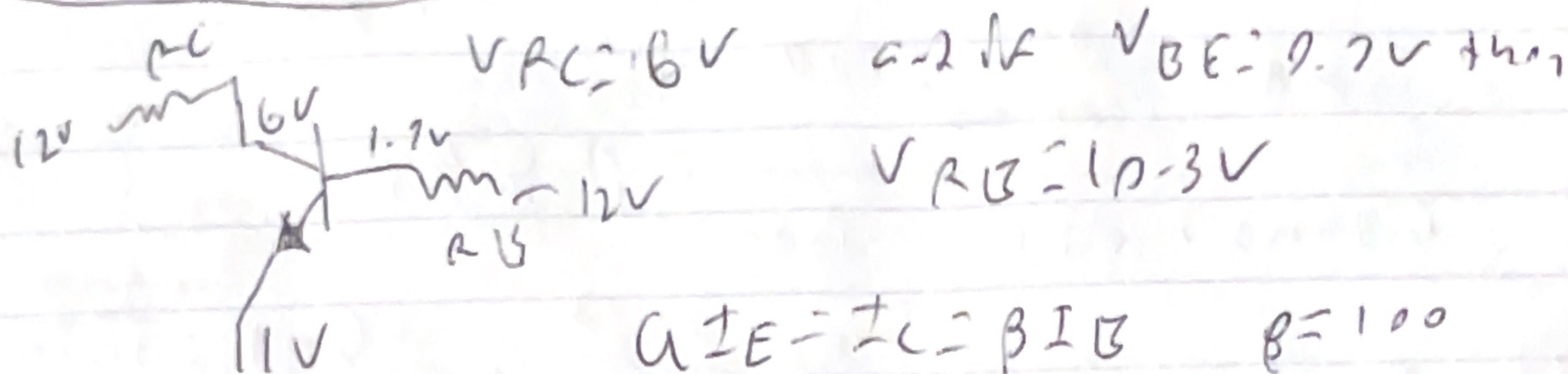
2.



Q point $\beta = 100$

26 $I_{RE} = 20 \text{ mA}$, $V_{RE} = 1 \text{ V}$, $V_{CE} = 5 \text{ V}$ $R_E, R_C, R_B = ?$

$R_E = \frac{V_{RE}}{I_{RE}} = 50 \Omega$ if $V_{CC} = 12 \text{ V}$ and $V_{CE} = 5 \text{ V}$ then



$\frac{\alpha}{1-\alpha} = \beta$ $\frac{\alpha}{1-\alpha} = 100 \Rightarrow \alpha \approx 0.99$

$I_{RE} = I_E = 10 \text{ mA}$ $I_C = 0.99 \cdot 10 \text{ mA} = 9.9 \text{ mA}$

$I_B = \frac{9.9 \text{ mA}}{100} = 99 \mu\text{A}$

$I_C = I_{RC}$ $I_B = I_{RB}$

$R_C = \frac{V_{RC}}{I_{RC}} = \frac{6}{9.9 \cdot 10^{-3}} = 606 \Omega$

$R_B = \frac{V_{RB}}{I_{RB}} = \frac{10.3}{99 \cdot 10^{-6}} = 104 \text{ k}\Omega$

27 $V_E, V_C, V_{CE}, I_C = ?$ (vins "removed")

$I_B R_B + V_{BE} + 2 I_E R_E = V_{CC}$

$0.99 I_E = 100 I_B$

$I_E = 101 I_B$

$I_B \cdot 10.4 \text{ k} + 0.7 + 100 I_E = 12$

$I_B \cdot 10.4 \text{ k} + 100 \cdot 101 I_B = 11.3$

$114 \text{ k} I_B = 11.3$

$I_B = 99 \mu\text{A}$

$V_E = 2 I_E \cdot \beta \cdot R_E$

$I_C = 9.9 \text{ mA}$

$V_E = 2 \cdot 9.9 \cdot 10^{-3} \cdot 100 \cdot 50 \Omega$

$V_E = 1 \text{ V}$

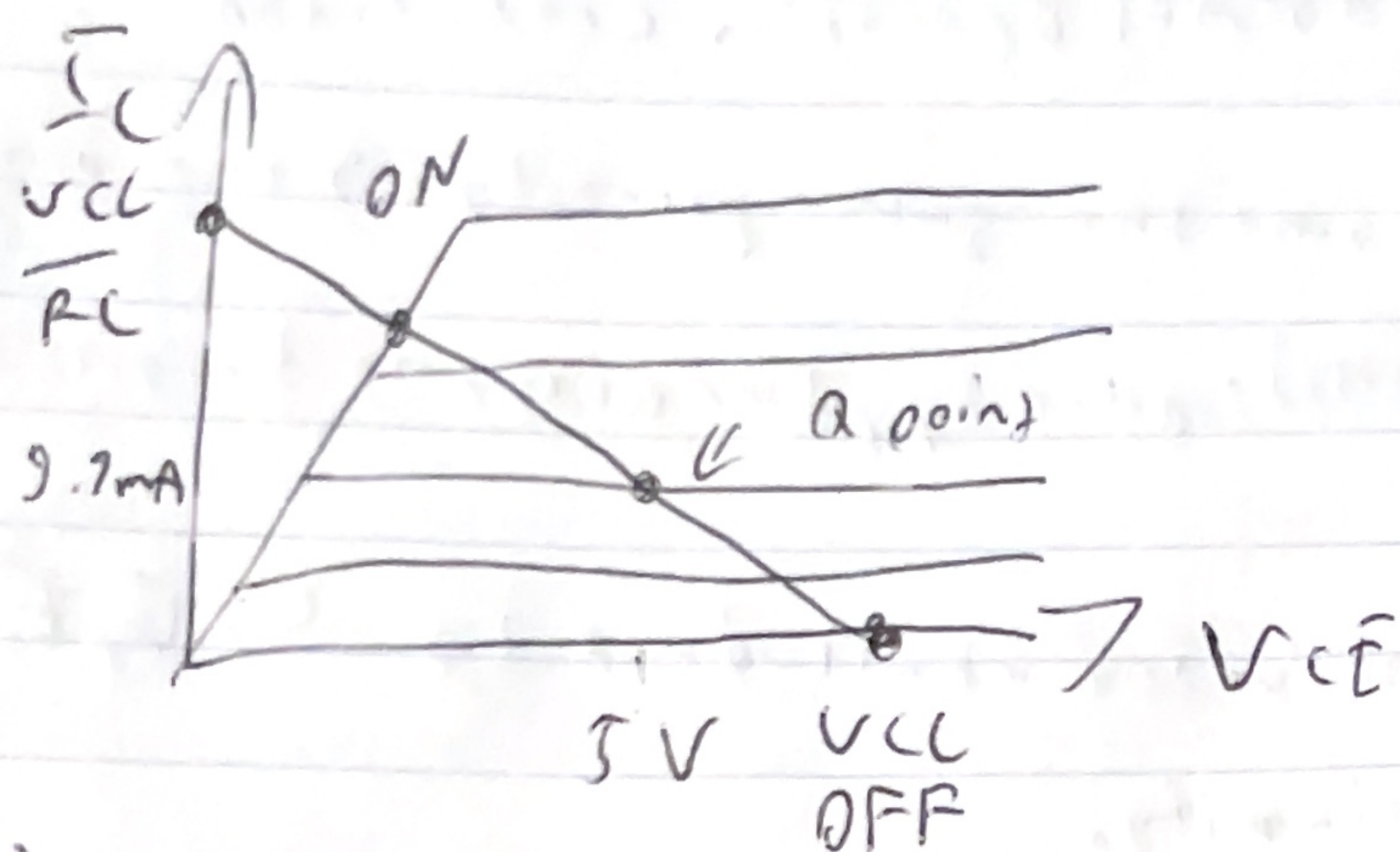
$V_C = V_{CC} - \beta I_B R_C$

$V_C = 6.7 \text{ V}$

$V_{CE} = V_C - V_E = 5.7 \text{ V}$

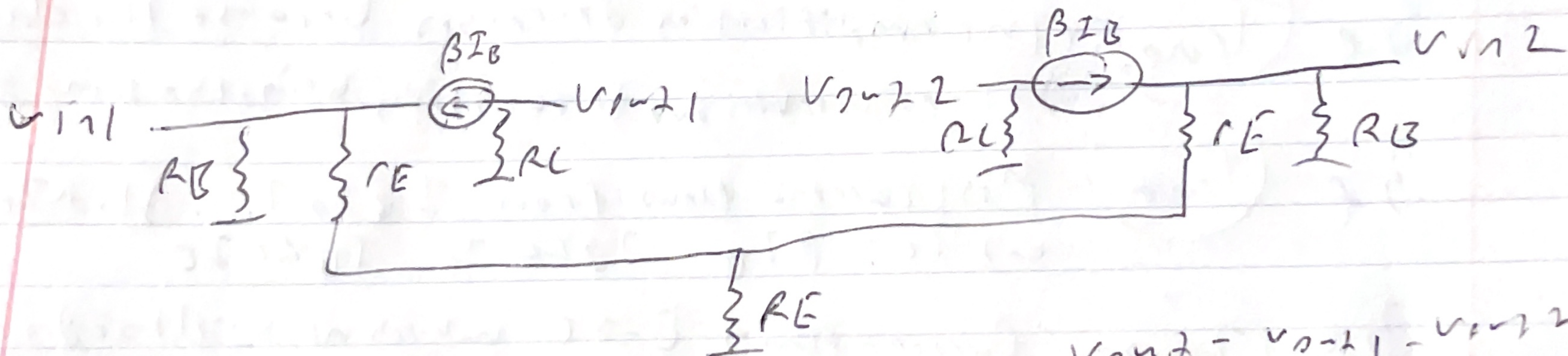
2c ~~conf~~ My V_C , V_E , and V_{CE} values are totally wrong. I think it's because of my resistor values... don't know how to calculate those correctly

Since I_{AE} is constant (due to symmetry of circuit), V_{AE} is constant, which makes V_E constant, or a virtual ground.



2d

$$r_E = \frac{V_T}{I_E} = \frac{26mV}{10mA} = 2.6 \Omega$$



2e $A_{VOL} = ? = \frac{v_{out}}{v_{in}}$

$v_{in1} = -v_{in2}$ $v_{out} = v_{out1} - v_{out2} = -2v_{out1}$

$$A_{VOL} = \frac{v_{out1} - v_{out2}}{2v_{in1}} = \frac{\beta I_{B1} R_C - \beta I_{B2} R_C}{2 \cdot 2 I_E r_E}$$

$$A_{VOL} = \frac{2 \beta I_{B1} R_C}{4 I_E r_E} \rightarrow \frac{\beta I_{B1} R_C}{2 I_E r_E} \rightarrow \beta I_B \approx I_E$$

$$\frac{R_C}{2 r_E} = A_{VOL}$$

$$2f \quad A_{vcl} = \frac{R_c}{r_{ie}} = \frac{60k}{2.2k} = \underline{116.5}$$

$$2g \quad A_{vcl} \cdot A_{voc} = 116.5 \cdot 116.5 = \underline{13572}$$

3a (True) $A_{IE} = I_c$. A is always less than 1, so I_c is always less I_E , in normal operation conditions.

3b (True) Base will become the same polarity as the emitter, and collector will be opposite polarity, turning the transistor into a type of diode.

3c (False) It will not work at all due to C, B, E all being the same polarity.

3d (True) C, B, E will all be the same polarity. It will not work.

3e (True) The amplification decreases because the circuit stops working due to everything being the same polarity.

3f (True) Most current flows from I_C to I_E . Also $I_E = I_C$ and $I_C = \beta I_B$ $I_B \ll I_C$ $I_B \ll I_E$

3g (True) Current flows $E \rightarrow C$ which means voltage drop is positive $C \rightarrow E$ so since $B \rightarrow C$ is backwards, or reverse biased.